

Integration Bee

Sample Problems

Questions

1

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

2

$$\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$$

3

$$\int \frac{dx}{x^4 + 1}$$

4

$$\int \left(\frac{\ln x}{x} \right)^2 dx$$

5

$$\int_0^{\infty} \frac{\sin^2 x}{x^2} dx$$

6

$$\int_0^{\infty} \frac{x}{(1+x^4)(1+x)} dx$$

7

$$\lim_{n \rightarrow \infty} n \int_0^{\pi/2} [1 - (\cos x)^{1/n}] dx$$

8

$$\int \frac{[x + \sqrt{1+x^2}]^n}{\sqrt{1+x^2}} dx$$

9

$$\int e^x (\cos 2x + \sin 2x) dx$$

10

$$\int_0^{\pi/2} \frac{dx}{\sin x + \sec x}$$

Hints

- 1 Substitute $t = \frac{1}{x}$.
- 2 Substitute $x = \tan \theta$ or use integral of inverse functions.
- 3 Try to factor $1 + x^4$.
- 4 Substitute $\ln x = t$ and do Integration by parts.
- 5 Use the Feynman technique.
- 6 Substitute $t = \frac{1}{x}$.
- 7 Use Taylor series.
- 8 Substitute $x = \tan z$.
- 9 Differentiate $e^x(A \sin(2x) + B \cos(2x))$ and equate coefficients.
- 10 Use King property and substitute $\sin x - \cos x = t$.

Answers

1 $\frac{\pi}{8}$

2 $\frac{\pi^2}{4}$

3 $\frac{1}{4\sqrt{2}} \ln \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + \frac{1}{2\sqrt{2}} \arctan \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C$

4 $-\frac{2 + 2 \log x + (\log x)^2}{x} + C$

5 $\frac{\pi}{2}$

6 $\frac{\pi}{8}$

7 $\frac{\pi}{2} \ln 2$

8 $\frac{[x + \sqrt{1 + x^2}]^n}{n} + C$

9 $\frac{e^x}{5} (3 \sin 2x - \cos 2x) + C$

10 $\frac{1}{\sqrt{3}} \ln(2 + \sqrt{3})$